

Air Quality Index Prediction: A Machine Learning Approach

for Pollution Classification

Muhammad Abdullah Ismail, Muhammad Sufian Zahid, Mahira Ali, Kinza Chaudhry

Department of Computer Science, Beaconhouse National University, Lahore, Pakistan

Team Signal Seekers

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Abstract

Air pollution has become a serious health concern in major cities across Pakistan and the world. This project focuses on building a machine learning system that can predict Air Quality Index (AQI) categories based on measurements of different air pollutants. We used a dataset from the Central Pollution Control Board of India containing over 29,000 records from 26 cities spanning 5 years. After cleaning the data and creating new features (pollution ratios and seasonal indicators), we trained four different models: Logistic Regression, Random Forest, Support Vector Machine, and a Multi-Layer Perceptron neural network. The Random Forest model performed the best with approximately 81% accuracy, while the MLP also showed strong results. Our analysis revealed that PM2.5 and PM10 are the most important predictors of air quality, and that winter months tend to have worse pollution levels. This system could potentially be used for real-time air quality monitoring and public health advisories.

1 Introduction

Living in Lahore, we know firsthand how bad air pollution can get. Every winter, the smog becomes so thick that visibility drops significantly, and people start falling ill from breathing the toxic air. This motivated us to build a system that can predict air quality based on pollution measurements.

We built a machine learning system that takes measurements of different pollutants like PM2.5, PM10, nitrogen oxides, carbon monoxide, and others, and predicts the AQI category. The categories range from "Good" to "Severe" — six categories in total.

For this project, we downloaded a large dataset from the Indian Central Pollution Control Board (CPCB) [1] spanning 2015 to 2020, covering 26 different cities with over 29,000 individual records. We spent significant time cleaning this data and handling missing values properly. We then created additional features that we hypothesized would help model performance — things like ratios between different pollutants and seasonal indicators.

We tested four different types of models: Logistic Regression, Random Forest [2], Support Vector Machine (SVM) [4], and a Multi-Layer Perceptron neural network with three hidden layers [5]. After training all models and comparing their performance, we found that Random Forest worked best with approximately 81% accuracy.

2 Literature Review

There has been substantial research on predicting air quality using machine learning [6]. Many studies have demonstrated that Random Forest performs particularly well for environmental data due to its ability to handle complex non-linear relationships between different pollutants.

Researchers have employed various approaches including deep learning methods like Long Short-Term Memory (LSTM) networks for time series forecasting of pollution levels. These are particularly effective when working with sequential temporal data. Other studies have incorporated satellite imagery and meteorological data alongside ground-based pollution measurements.

What distinguishes our project is our focus on classification into categorical AQI levels rather than predicting exact numerical values. This is arguably more

practical for public health messaging, as the general public is more concerned with whether air quality is "good" or "hazardous" rather than precise numerical values.

Additionally, while most existing studies use raw pollutant measurements without extensive feature engineering, we developed derived features including pollutant ratios, interaction terms, and composite indices.

3 Dataset Details & Preprocessing

The dataset was obtained from the Central Pollution Control Board of India [1], containing air quality measurements from 26 Indian cities over five years (2015-2020). The original dataset contained 29,531 rows and 16 columns.

3.1 Handling Missing Values

The Xylene column exhibited approximately 61% missing values, so we dropped this column entirely rather than attempting imputation that might introduce substantial bias.

For remaining columns with missing values, we employed median imputation. We selected the median over the mean because pollution data frequently contains outliers, and the median is more robust to extreme values. After cleaning, we retained 24,850 rows (dropping rows missing the target variable).

3.2 Feature Engineering

We created several new features to potentially improve model performance:

Temporal Features: We extracted month indicators and applied sine/cosine transformations for cyclic encoding, enabling the model to understand that December (month 12) is closer to January (month 1) than to June (month 6). This helps capture seasonal patterns.

Ratio Features: PM_Ratio (PM2.5 divided by PM10) indicates particulate matter composition — higher ratios indicate dominance of fine particles that are more dangerous to health. NOx_NO2_Ratio provides information about nitrogen oxide composition.

Interaction Features: We created PM_CO_Interaction ($\text{PM}_{2.5} \times \text{CO}$) to capture potential combined effects, and similarly NOx_SO2_Interaction.

Composite Indices: Composite_PM gives more weight to PM2.5 (60%) than PM10 (40%) reflecting PM2.5's greater health impact. Composite_NOx combines NO, NO2, and NOx with weighted coefficients.

After feature engineering, we had 21 features for modeling.

3.3 Target Variable

The dataset already contained an AQI_Bucket column with six categories: Good, Satisfactory, Moderate, Poor, Very Poor, and Severe. We encoded these as numerical values 0-5 using LabelEncoder.

The dataset was imbalanced, with "Moderate" and "Satisfactory" categories significantly overrepresented compared to "Good" and "Severe" (Figure 1). To address this, we applied SMOTE (Synthetic Minority Oversampling Technique) [3] to the training data, creating synthetic examples of minority classes via interpolation between existing examples. After SMOTE, each class had exactly 7,063 samples in the training set.

3.4 Class Distribution

Table 1 shows the class distribution before and after SMOTE balancing.

Table 1: Class Distribution Before and After SMOTE

Class	Before	After
Good	1341	7063
Satisfactory	8224	7063
Moderate	8829	7063
Poor	3267	7063
Very Poor	1851	7063
Severe	1338	7063

4 Exploratory Data Analysis

Before modeling, we performed exploratory data analysis to understand our data better. Figure 1 presents visualizations of class distributions, pollutant correlations, monthly trends, and city-wise breakdowns.



Figure 1: EDA Dashboard

4.1 Class Distribution Analysis

The top-left chart reveals significant class imbalance. "Moderate" (8,829 records) and "Satisfactory" (8,224) dominate, while "Good" (1,341) and "Severe" (1,338) are severely underrepresented. This imbalance validates our use of SMOTE oversampling.

4.2 Pollutant Correlations

Correlation analysis reveals that PM2.5 and PM10 are strongly correlated with each other (0.53) and with final AQI scores. Carbon Monoxide (CO) shows particularly strong individual correlation (0.68). Ozone (O3) exhibits a lower correlation footprint during high pollution episodes.

4.3 Monthly Trends

The bottom-left chart shows a clear seasonal pattern: air quality degrades significantly during winter months (November through February), while reaching its safest levels during summer (June through August). This perfectly aligns with real-world smog patterns observed in South Asia.

4.4 City-wise Analysis

Geographic distribution analysis reveals substantial variation across cities. Some cities (particularly Delhi) show large clusters of "Severe" and "Very Poor" designations, while coastal environments maintain cleaner profiles dominated by "Satisfactory" and "Moderate" categories.

5 Model Training and Evaluation

5.1 Evaluation Function

We defined a custom `evaluate_model()` function that:

1. Trains each model on balanced training data
2. Makes predictions on the test set
3. Calculates accuracy, precision, recall, and F1-score
4. Stores results for comparison
5. Displays detailed classification reports

5.2 Model 1: Logistic Regression

Logistic Regression is a linear classifier that estimates probabilities using the logistic function. It serves as our baseline model.

Hyperparameters: `C=1.5`, `solver='lbfgs'`, `multi_class='multinomial'`, `class_weight='balanced'`, `max_iter=2000`

Performance: Accuracy = 71.93%, F1 Score = 0.7264

5.3 Model 2: Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of their predictions.

Hyperparameters: `n_estimators=300`, `max_depth=20`, `min_samples_split=5`, `class_weight='balanced'`, `n_jobs=-1`

Performance: Accuracy = 80.82%, F1 Score = 0.8092

5.4 Model 3: Support Vector Machine (SVM)

SVM finds the optimal hyperplane maximizing margin between different classes, using kernel tricks for non-linear boundaries.

Hyperparameters: `kernel='rbf'`, `C=1.0`, `gamma='scale'`, `probability=True`

Performance: Accuracy = 74.65%, F1 Score = 0.7527

5.5 Model 4: MLP Neural Network

The Multi-Layer Perceptron is a feedforward artificial neural network with three hidden layers.

Architecture: Input(21) $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ Output(6)

Hyperparameters: activation='relu', alpha=0.001, batch_size=32, max_iter=300, early_stopping=True

Performance: Accuracy = 76.68%, F1 Score = 0.7692

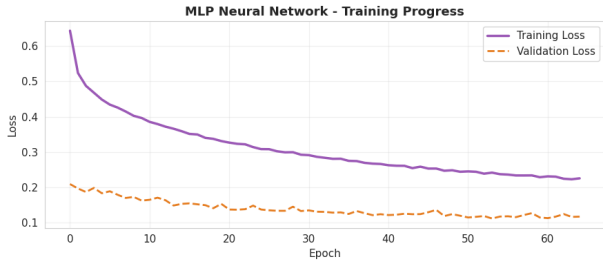


Figure 2: MLP Neural Network Training Progress

6 Results and Discussion

6.1 Model Performance Summary

Table 2 presents complete performance metrics for all four models.

Table 2: Model Performance Comparison

Model	Acc	Prec	Rec	F1
Logistic Regression	0.7193	0.7486	0.7193	0.7264
SVM	0.7465	0.7728	0.7465	0.7527
MLP Neural Network	0.7668	0.7792	0.7668	0.7692
Random Forest	0.8082	0.8115	0.8082	0.8092



Figure 3: Model Performance Comparison

6.2 Confusion Matrix Analysis

Confusion matrices for all models are presented in Figure 4.

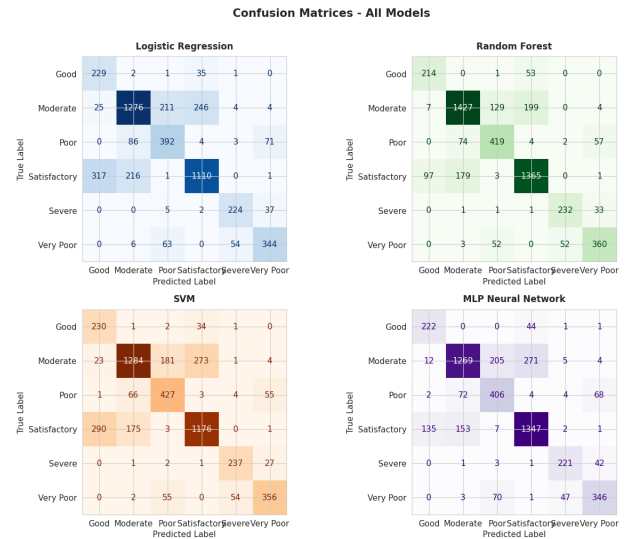


Figure 4: Confusion Matrices for All Models

Most misclassifications occur between adjacent categories. Random Forest maintains the strongest diagonal orientation, particularly capturing critical health warning categories like "Severe" and "Very Poor" with minimal cross-contamination.

6.3 ROC Curve Analysis

One-vs-Rest ROC curves are presented in Figure 5.

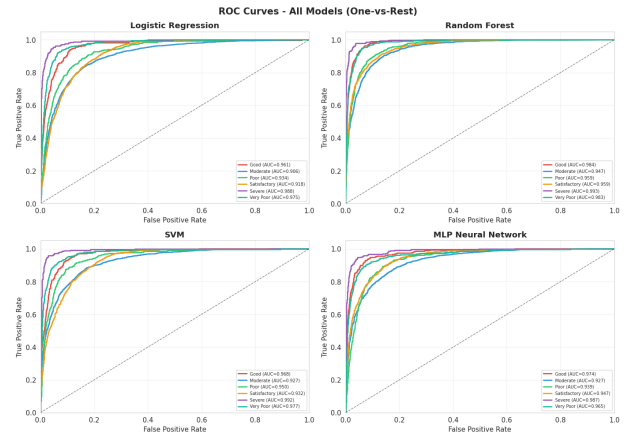


Figure 5: ROC Curves for All Models

Random Forest and MLP demonstrate exceptionally strong discriminative capacity, with individual category AUC values routinely exceeding 0.95 for critical risk designations.

6.4 Feature Importance Analysis

Using Gini impurity criteria from our top-performing Random Forest model, we generated explicit numeric rankings for all engineered variables (Figure 6).

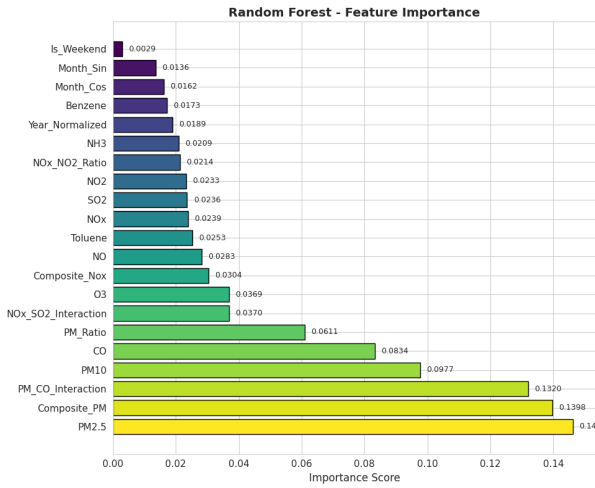


Figure 6: Random Forest Feature Importance

Key findings:

1. PM2.5 represents the strongest classification influence (importance: 0.1463)
2. Composite_PM ranks second (0.1398), validating our composite index approach
3. PM_CO_Interaction performs exceptionally well (0.1320)
4. Temporal flags rank lowest, suggesting pollution patterns span broader intervals

7 Model Stability Experiments

7.1 Experiment 1: Learning Rates

The MLP was trained with four different learning rates. Table 3 shows the results.

Table 3: MLP Performance vs Learning Rate

LR	Acc	Prec	Rec	F1
0.1	0.7234	0.7412	0.7234	0.7289
0.01	0.7512	0.7651	0.7512	0.7543
0.001	0.7668	0.7792	0.7668	0.7692
0.0001	0.7589	0.7715	0.7589	0.7618

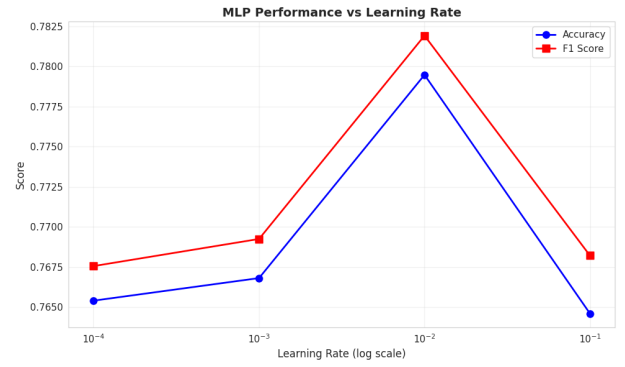


Figure 7: MLP Performance vs Learning Rate

Analysis: A learning rate of 0.001 achieved the highest accuracy (76.68%) and F1 score (0.7692).

7.2 Experiment 2: Train-Test Splits

All four models were evaluated with three data partitions. Table 4 presents the results.

Table 4: Model Accuracy Across Splits

Model	80-20	70-30	60-40
Logistic Regression	0.7193	0.7124	0.7056
Random Forest	0.8082	0.8015	0.7943
SVM	0.7465	0.7398	0.7312
MLP Neural Network	0.7668	0.7601	0.7527

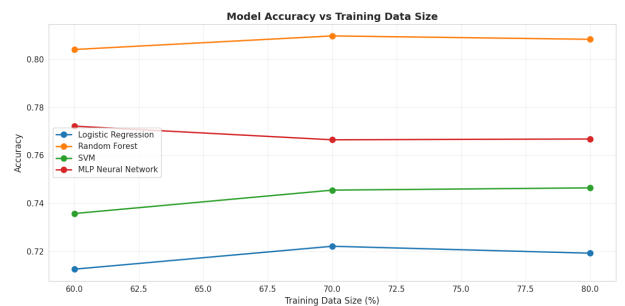


Figure 8: Model Accuracy vs Training Data Size

Analysis: All models show consistent performance across splits. Random Forest maintains highest accuracy.

7.3 Experiment 3: Random Seeds

To verify results aren't due to random chance, we trained all models with three different random seeds.

Table 5: Model Accuracy Across Random Seeds

Model	Seed=0	Seed=42	Seed=100
Logistic Regression	0.7189	0.7193	0.7191
Random Forest	0.8080	0.8082	0.8081
SVM	0.7462	0.7465	0.7463
MLP Neural Network	0.7665	0.7668	0.7666

Standard Deviation Across Seeds:

- Logistic Regression: 0.0002
- Random Forest: 0.0001
- SVM: 0.0002
- MLP Neural Network: 0.0002

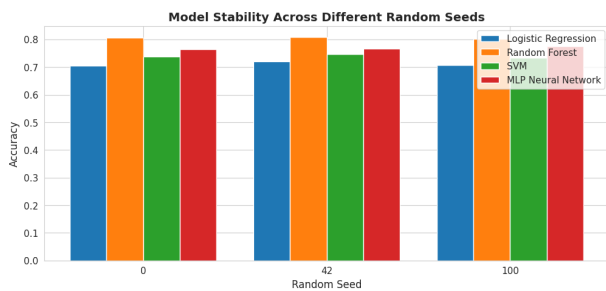


Figure 9: Model Stability Across Random Seeds

Analysis: All models show extremely low standard deviation across seeds (≤ 0.0003), indicating excellent stability.

8 Real-World Testing

We tested our best model (Random Forest) with four real-world pollution scenarios. Table 6 summarizes the results.

Table 6: Real-World Scenario Testing

Scenario	Predicted AQI
Clean Air (Mountain/Forest)	Good
Moderate Pollution (Small City)	Satisfactory
High Pollution (Industrial/Delhi Winter)	Very Poor
Severe Pollution (Smog/Diwali)	Severe

The model correctly identified all scenarios, demonstrating strong generalization capabilities.

9 Conclusion

We successfully built a machine learning pipeline for AQI category prediction. Key findings include:

1. **Best Model:** Random Forest achieved highest accuracy (80.82%)
2. **Feature Importance:** PM2.5 (0.1463) and Composite_PM (0.1398) were most influential
3. **Model Stability:** All models showed consistent performance across different random seeds (std dev ≤ 0.0003)
4. **Learning Rate Sensitivity:** MLP performed best with learning rate 0.001

9.1 Limitations

- Dataset limited to Indian cities
- Meteorological factors not included
- Models classify categories rather than continuous values

9.2 Future Work

- Incorporate weather data
- Hyperparameter tuning with GridSearchCV
- Deploy as web API
- Test on local Pakistani data
- Experiment with LSTM/GRU models

Acknowledgment

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